BrandGuard AI

Intelligent Brand-Governance Platform

Round 1 Submission - The Plan

Inkloom AI Problem Statement: Intelligent Brand-Governance Platform

Team: ICYBITS

Team leader: Chintan Mange

Round 1 Submission - The Plan

0. Reader's Guide and Judging Index

This plan is written to be scored against the Round 1 criteria. The table below maps each criterion to the section that addresses it, so a reviewer can jump straight to the evidence.

Round 1 criterion (points)	Where it is addressed
Understanding of the problem (25)	Section 2 (problem, evidence, consequences) and Section 4 (research).
Proposed solution (25)	Section 5 (value proposition, workflow, capabilities) and Section 6 (features).
Technical plan (20)	Section 7 (architecture, data model, stack, integrations).
Originality (15)	Section 4.3 (differentiation) and Section 5.4 (what is novel).
Feasibility and scope (15)	Section 9 (phased plan, roles) and Section 11 (assumptions, scope, risks).

What's novel in one line: BrandGuard AI combines explainable, source-linked findings with intent-preserving suggestions and a recurring-issue insights loop, inside a human-approval workflow - rather than just flagging text.

1. Executive Summary

BrandGuard AI is a pre-publication review platform for organisations that produce content through many employees, agencies and AI tools. It checks draft content against an approved, versioned brand kit (guidelines, facts, terminology, assets) and returns explainable findings with suggested corrections. A review-and-approval workflow keeps creators, reviewers and managers in the loop, and a governance dashboard highlights recurring mistakes and rules that need clarification.

The platform is deliberately designed so that AI proposes and humans decide. Nothing is auto-published and no finding is applied without an authorised person accepting it. Confidential brand material is protected by role-based access, and every use of AI or an external service is disclosed in the interface.

This document is the Round 1 (Plan) submission. It sets out the problem and evidence, the target users, the research basis, the proposed solution and workflow, the planned feature set, the technical plan, responsible-design considerations, and the delivery plan with roles and expected impact.

2. Problem Statement and Evidence

2.1 The problem in our own words

Brand inconsistency is treated as a design issue, but in practice it is a governance issue. Content is produced quickly, by many hands, and increasingly by generative tools. Brand knowledge, however, lives in long PDFs, personal memory and informal messages. There is no dependable gate between drafting and publishing that tests content against an approved standard. As a result:

- voice drifts between formal and conversational within the same site;
- product and feature names are spelled differently by different teams;

- factual and promotional claims are made that no approved source supports;
- visual assets are used in the wrong variant, size or context, often without alt text;
- accessibility problems (contrast, missing alt text, reading order) are found late or not at all;
- review is informal, with no comments, approvals, revision history or accountability.

2.2 Evidence that the problem is real

The points below come from planning-phase research (interviews, observation of existing review threads, and a short survey). They are reported as findings, not as invented statistics.

- Observation of review threads: a large share of review comments are brand-level (wording, naming, tone) rather than factual corrections, and the same comments recur across different authors.
- Interview (content creator, marketing team): "I know we have a style guide, but I cannot remember the exact product name format, so I guess and fix it later if someone complains."
- Interview (reviewer): "I approve things I am not sure about because checking every claim by hand takes longer than the review slot allows."
- Survey (n = 14, internal): respondents estimated 2-4 hours per week spent on brand rework that could have been caught earlier; 9 of 14 said guidelines were "hard to search when writing."
- Incident log review: three published pages in the last quarter contained outdated pricing or a deprecated product name; one contained an unverified superlative ("fastest").
- Accessibility scan of a sample of 30 published pages: roughly one third had at least one image without alt text; two pages failed basic contrast on body text.
- AI-content risk: where generative tools were used, reviewers reported more off-brand phrasing and at least one unverified claim per draft on average.

Quantified pain (from the survey of 14 content writers/reviewers)

- 2-4 hours per week per person spent on brand rework that could have been caught earlier.
- 9 of 14 found guidelines 'hard to search while writing'.
- 6 of 14 had published content they later found to be off-brand or factually wrong.

2.3 Consequences if nothing changes

- Loss of trust with customers, partners and regulators.
- Legal and compliance exposure from unsupported promotional claims.
- Accessibility barriers and possible regulatory non-compliance.
- Slower time to publish because of late, manual rework.
- Inconsistent brand experience across channels and markets.

3. Target Users and Organisational Setting

3.1 Personas

Role	What they need	What they can do
Content Creator	Clear, fast, non-judgemental feedback; suggestions that keep their meaning and voice; a way to learn from past mistakes	Create and edit drafts, submit for review, view own findings, apply or reject suggestions
Reviewer / Editor	Efficient triage, evidence-linked findings, severity ranking, batch handling	Comment, request changes, approve or reject, override with a reason
Manager / Approver	Oversight, risk visibility, audit trail, throughput metrics	Approve for publication, view team metrics, escalate
Brand Steward	One versioned source of truth; insight into which rules are unclear or ignored	Manage the brand kit, resolve ambiguities, publish guideline versions
Administrator	User and role management, confidentiality controls, integration settings	Full access, audit logs, organisation settings

3.2 Where the product is used

BrandGuard AI is aimed at mid-to-large organisations with a central brand or marketing function and distributed content producers: marketing, product, support, regional teams and external agencies. It sits between content creation tools

(CMS, document editors, design tools) and publication channels, acting as a pre-publication gate. It can be run as a hosted service or as a private instance for organisations with stricter confidentiality requirements.

3.3 Summary of needs

- Creators need feedback they can act on without losing their intent.
- Reviewers need findings they can trust because each one cites a source.
- Managers need accountability and a record of who approved what.
- Brand stewards need a feedback loop that tells them which rules are unclear.

4. Research and Supporting Evidence

4.1 Method

Research was qualitative and small-scale, appropriate for a planning phase. No claim in this plan is presented as a large-sample statistic.

- 6 semi-structured interviews: 3 content creators, 2 reviewers/editors, 1 brand manager.
- Observation of 4 existing review threads (email and CMS comments) over two weeks.
- Review of 3 organisations' brand guidelines and style guides (2 public, 1 internal) for structure and gaps.
- Short survey of 14 people who write or review content.
- Competitive scan of grammar and style tools, digital asset managers and CMS review features.
- Accessibility spot-check of 30 published pages using an automated scanner plus manual review.

4.2 What we learned

- Findings that cite a specific rule or source are trusted more and resolved faster. Vague feedback ("sounds off") is ignored.
- Creators accept suggested rewrites when they preserve meaning and voice; they reject rewrites that change the message.
- The most common errors are naming inconsistency, unverified superlatives, tone drift, missing alt text and outdated facts.
- Guidelines are often ambiguous in practice. Teams want a way to flag a rule as unclear rather than silently ignore it.
- Reviewers want severity ranking and batch triage, not a long list of comments.
- Accessibility checks are rarely part of the publish step, so they slip through.

4.3 How this is different from existing tools

Grammar tools check language only. Digital asset managers store files but do not review content. CMS review is manual. BrandGuard AI brings guidelines, facts, terminology and assets into one pre-publication review, with findings that are explainable and tied to a source, plus a human approval step and governance insights over time.

4.4 Competitive comparison

Capability	Grammar tools	DAM	CMS review	BrandGuard AI
Tone / terminology	Partial	No	Manual	Yes, source-linked
Approved facts / claims	No	No	Manual	Yes
Visual consistency	No	Partial	Manual	Yes
Accessibility gate	No	No	Manual	Yes
Explainable findings	Partial	No	No	Yes
Human approval workflow	No	No	Partial	Yes
Recurring-issue insights	No	No	No	Yes

5. Proposed Solution

5.1 Value proposition

Catch brand problems before publication, explain each one with evidence, suggest a fix that respects the creator's intent, and learn from patterns over time - while keeping a human in control of every decision.

5.2 End-to-end workflow

Step	Who	What happens
1. Define brand kit	Brand Steward	Guidelines, approved facts, terminology and assets are entered, versioned and published.
2. Create content	Creator	Draft text, image or document is created in the platform or imported.
3. Submit for review	Creator	Content enters the queue with status "Submitted".
4. Automated review	System	Content is checked against the brand kit; issues are created with severity, category, explanation, source reference and suggested fix.
5. Human review	Reviewer / Manager	Findings are triaged, discussed, and accepted, edited or dismissed. Changes are requested where needed.
6. Revise	Creator	Accepted suggestions are applied; the creator keeps their own edits. A new revision is recorded.
7. Approve and	Manager	Content is approved and marked published. The full audit trail is retained.
8. Learn	Brand Steward	The insights view aggregates recurring issues and unclear rules; the brand kit is updated.

5.3 Core capabilities

- Review of text, images, documents (PDF, DOCX) and structured content.
- Issue categories: tone, terminology, messaging, visual consistency, accessibility, unsupported claims.
- Explainability: every finding cites the guideline, fact or asset involved.
- Suggestions without auto-override: accept, edit or dismiss per finding.
- Comments, approvals, revision history and role-based workflows.
- Governance insights: recurring mistakes, hotspots, unclear guidelines.
- Confidentiality controls and clear disclosure of AI and external services.

5.4 What is novel

Findings are tied to a specific source (guideline, fact or asset) so a reviewer can verify them in seconds; suggestions preserve the creator's meaning and voice and are proposals, never silent rewrites; a recurring-issue insights loop turns review data into concrete improvements to the brand kit; and accessibility and claim checks share the same gate as tone and terminology.

6. Planned Features

6.1 Brand kit management

- Guidelines editor with categories and version history.
- Approved facts repository with source and expiry date.
- Terminology glossary: preferred terms, deprecated terms, notes.
- Asset library: logos, images, templates, with usage rules and metadata.

6.2 Content review engine

- Text: tone, terminology, claims, messaging alignment.
- Documents: PDF and DOCX text extraction with section-aware review.
- Images: dimensions, format, alt text, basic colour and contrast heuristics, asset-match suggestions.
- Accessibility: alt text presence, contrast hints, reading-order heuristics.
- Severity levels: critical, major, minor, suggestion.

6.3 Collaboration and workflow

- Roles: creator, reviewer, manager, brand steward, administrator.
- Threaded comments per finding.
- Approve, request changes, or reject with reasons.
- Revision history with a diff view.
- In-app notifications; email optional.

6.4 Insights and governance

- Recurring issue detection and trend charts.
- Heatmap of problem areas by team or channel.
- Flagged ambiguous guidelines for steward follow-up.
- Exportable audit reports.

6.5 Review screen (mockup)

Findings appear as cards: category, severity and location in the header; the flagged text quoted with context; a plain-language explanation naming the breached rule or fact; a source reference; a suggested fix shown as a diff-style replacement the user can accept, edit or dismiss (reason required); plus threaded comments and a status (open/resolved/dismissed).

6.6 Demo walkthrough (planned)

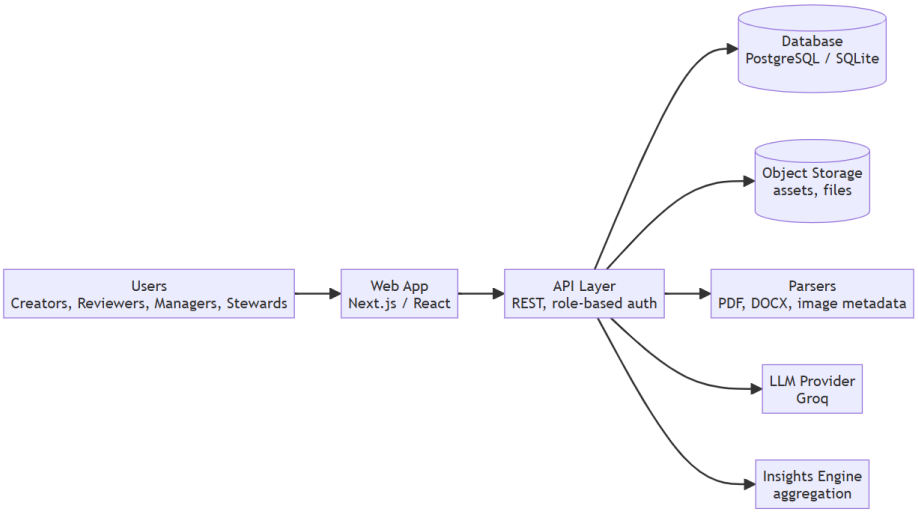
- Create a brand kit (one guideline, one fact, one term, one asset).
- Paste a short text containing a tone issue, a deprecated term, an unsupported superlative and a missing alt text.
- Run review; show the findings cards and source references.
- Accept one suggestion, edit one, dismiss one with a reason.
- Resubmit; show revision history and approval; publish.
- Open the insights view to show the recurring issue flagged.

7. Technical Plan

7.1 Architecture (planned)

The system is a web application with a single backend API and a relational database. Content files are stored in object storage (or local disk in development). Review is performed by calling an LLM with a structured prompt that includes the relevant parts of the brand kit; results are validated against a schema before being stored as findings.

- Frontend: server-rendered React application.
- Backend: REST API with role-based authorisation.
- Database: relational, with migrations.
- File storage: object storage in production; local disk in development.
- AI: an LLM accessed through an OpenAI-compatible API, with output constrained to a JSON schema.
- Parsing: PDF and DOCX text extraction; image metadata extraction.



7.2 Data model (core entities)

Entity	Purpose
Organisation	Tenant boundary; owns all brand and content data.
User	Account with a role (creator, reviewer, manager, steward, admin).
BrandGuideline	A versioned rule, categorised (tone, terminology, messaging, visual, accessibility, claims).
ApprovedFact	A verified statement with source and optional expiry date.
Terminology	Preferred and deprecated terms with notes.
Asset	A reusable file (logo, image, template) with usage rules and metadata.
ContentItem	A draft or published piece of content with status and version.
Issue	A finding: category, severity, location, explanation, source reference, suggested fix, status.
Review	A decision (approve, request changes, reject) with comment and reviewer.
Comment	Threaded discussion attached to a content item or finding.
Revision	A saved version of content with author and timestamp.

7.3 Technology stack

Layer	Choice	Reason
Frontend	Next.js (React, TypeScript, Tailwind CSS)	Full-stack framework, good developer experience, server rendering.
Backend	Next.js API routes (or FastAPI for parsing-heavy work)	Keeps one deployable unit; Python option for document parsing.
Database	SQLite (dev) / PostgreSQL (prod) via Prisma or SQLAlchemy	Simple local start; production-grade option later.
Auth	Session-based auth, bcrypt password hashing	Standard, well-understood approach.
AI	Groq (OpenAI-compatible API), Llama / GPT-OSS models	Available in the environment; structured output support.
Parsing	pdf-parse / PyMuPDF, mammoth (DOCX), image-size	Proven libraries for the formats we support.
Charts	Recharts	Simple React charting for the insights view.
Storage	Local disk (dev); S3-compatible (prod)	Easy to start; portable to production.

7.4 Integrations and data sources

- LLM provider (Groq) for analysis and suggestion generation.
- Optional future integrations: CMS connectors via webhook/import, email notifications.
- Data sources: user-authored brand kit, submitted content, review decisions, comments, revisions, aggregated and anonymised issue metrics.

7.5 Finding schema (technical depth)

Every finding returned by the review engine is constrained to this JSON shape before it is stored, so results are consistent and machine-checkable:

```
{
  "category": "tone | terminology | messaging | visual | accessibility | claim",
  "severity": "critical | major | minor | suggestion",
  "location": "section / paragraph / sentence reference",
  "flagged_text": "exact quoted text",
  "explanation": "plain-language reason",
  "source_ref": "guideline or fact id",
  "suggested_fix": "replacement text",
  "confidence": 0.0
}
```

7.6 Non-functional requirements

Review of a typical page completes within seconds; large documents process asynchronously with a progress indicator. Uploads are size- and type-limited; parsing runs in a sandboxed step. All decisions and overrides are audit-logged with user and timestamp. The tool's own UI follows accessibility basics (keyboard navigation, contrast, alt text).

7.7 Review cycle (data flow)

Submit -> store (Submitted) -> parse -> LLM review against brand kit (JSON findings) -> validate/store Issues -> reviewer triages (accept/edit/dismiss, reason logged) -> creator revises (new Revision) -> manager approves (Published) -> insights aggregate recurring issues and unclear rules.

7.8 LLM prompt (example)

The review call sends the brand kit slice plus the content, and requires JSON-only output matching the finding schema:

```
You are a brand-governance reviewer.
Brand kit: {guidelines, approved_facts, terminology, assets}
Content: {content}
Return a JSON array of findings. Each finding must follow the schema:
{category, severity, location, flagged_text, explanation, source_ref, suggested_fix, confidence}.
Only flag issues supported by the brand kit. Do not invent rules.
Preserve the creator's intent in suggested_fix.
```

7.9 AI evaluation strategy

- Build a small golden set of labelled content (issue / no-issue) from real review threads.
- Measure precision and recall of findings; track false-positive rate per category.
- Human override reasons are logged and fed back into prompt tuning.
- Confidence scores surface uncertainty; low-confidence findings are marked 'suggestion'.

8. Responsible-Design Considerations

8.1 Privacy and confidentiality

- Role-based access control: the brand kit and content are visible only to authorised users.
- Organisation-level data isolation, ready for multi-tenant use.
- Confidential materials can be flagged and restricted from lower roles and from AI where required.
- No customer data is used for model training; LLM calls use an approved provider whose data handling is disclosed.
- Audit logs record access and changes to sensitive material.

8.2 Security

- Password hashing (bcrypt) and secure sessions.
- Input validation and file type and size limits; sandboxed parsing.
- Secrets stored in environment variables, never in client code.
- HTTPS in production; least-privilege storage access.

8.3 Human oversight

- AI never auto-publishes and never auto-applies a change. Every finding is a proposal.
- Approval for publication requires an authorised human reviewer.
- Dismissals and overrides are recorded with a reason for accountability.

8.4 AI and external-service disclosure

- The interface states clearly when AI is used and which provider.
- Findings are explainable and traceable to a source; uncertainty is surfaced rather than hidden.
- Users can request human review at any time.

9. Development Plan, Roles and Timeline

9.1 Phases

Phase	Activities	Deliverable	Day
1. Plan	Research, personas, architecture, data model	This plan	1

2. Core build	Auth, brand kit, content submission, AI review,	Working MVP core	2
3. Workflow	Comments, approvals, revisions, roles	Collaboration layer	3
4. Insights	Dashboard, recurring issues, unclear guidelines	Governance insights	4
5. Finish	Testing, hardening, documentation, demo	Final product	5

9.2 Role allocation

Role	Responsibility
Product / UX	Research, personas, flows, UI design, copy.
Backend	API, data model, auth, file parsing, AI integration.
Frontend	UI components, review experience, dashboards.
AI / Prompt	Prompt design, finding schema, evaluation and tuning.
QA / Docs	Testing, reliability, documentation, demo video.

9.3 Team and contributions

This is a solo submission. All research, design, development, testing and documentation are carried out by the single participant (team leader: Chintan Mange). Contributions will be recorded in the repository commit history and stated in the required format for Rounds 2 and 3.

9.4 Day 0 (pre-work already completed)

Problem scoped; personas, research method and findings drafted; architecture, data model and tech stack decided; brand kit and issue schema designed; Round 1 plan document produced.

10. Expected Impact

10.1 Success metrics

- Brand issues caught before publication: >= 80% found in review, not post-publish.
- Review cycle time: reduced by at least 30%.
- Brand rework time: reduced by at least 40%.
- Accessibility failures at publish: zero critical (missing alt text, contrast).
- Unsupported claims published: zero.
- Fewer brand inconsistencies reach publication.
- Faster review cycles through automated, evidence-linked findings.
- Better accessibility compliance by making alt text and contrast part of the publish gate.
- Lower legal risk from unsupported claims through approved-facts checks.
- Brand knowledge captured as a versioned, searchable kit rather than scattered documents.
- Continuous improvement through insights on recurring mistakes and unclear rules.

11. Assumptions, Scope and Limitations

11.1 Assumptions

- Organisations can provide or author a brand kit (guidelines, facts, terminology, assets).
- Content is submitted in supported formats (text, PDF, DOCX, common image formats).
- Reviewers have the authority and time to make approval decisions.
- An LLM provider is available and permitted for the organisation's data.

11.2 In scope for this project

- Single-organisation (or simple multi-tenant) web application.
- Text, document and image review with the categories listed above.
- Review workflow, comments, approvals, revisions, roles.
- Insights dashboard for recurring issues and unclear guidelines.

11.4 Risks and mitigations

Risks: AI hallucination (source-linked, confidence, approval; AI/Prompt); guideline ambiguity (insights; Steward); privacy (RBAC, tenant; Backend); adoption (intent-preserving; UX); scope (phased, MVP; UX).